

FTC StarterBot TeleOp Student Workbook

Part 8 Drive Intake and Telemetry

Name _____ Team _____ Date _____

In this capstone lesson, you will trace one gamepad through the StarterBot program, calculate the commands sent to five devices, and test your predictions while following the POWERING ON safety sequence.

Learning targets

- ☐ I can match Java variables to configured devices.
- ☐ I can calculate left and right drive power.
- ☐ I can calculate intake power from two triggers.
- ☐ I can explain why a motor or servo is reversed.
- ☐ I can use telemetry and observation as evidence.
- ☐ I can perform the POWERING ON sequence correctly.

Before the video

Predict the three robot jobs controlled by Gamepad 1.

1. _____
2. _____
3. _____

Vocabulary notes

TERM	MY MEANING OR EXAMPLE
iterative OpMode	
hardwareMap	
split arcade	
CRServo	
telemetry	

Video focus notes

TIME	FOCUS	MY NOTES
0:09	One gamepad controls these three jobs	
0:19	The method that repeats after START	
0:31	What must match exactly	
0:42	Why one drive motor is reversed	
0:53	What happens after POWERING ON and before START	
1:14	Left stick job and right stick job	
1:26	Left and right arcade-drive equations	
1:39	Intake trigger equation	
1:48	Three devices receiving intakePower	
1:56	What telemetry helps us compare	

Hardware map audit

Write PASS only when the Java variable, quoted configuration name, and physical device all agree.

JAVA VARIABLE	QUOTED NAME	PHYSICAL DEVICE	PASS
leftDrive	left_drive	Left drivetrain motor	
rightDrive	right_drive	Right drivetrain motor	
intake	intake	Intake roller motor	
leftIntakeServo	left_intake_servo	Left corner feeder	
rightIntakeServo	right_intake_servo	Right corner feeder	

Lifecycle order

Number these program events from 1 through 6.

- ☐ loop repeats and reads the controls
- ☐ the runtime calls init
- ☐ the team presses STOP
- ☐ hardwareMap connects the devices
- ☐ the team presses START
- ☐ the runtime calls stop

Direction and BRAKE

Why might the right drive motor be reversed in code?

Complete the statement: BRAKE reduces _____ when zero power is commanded. It does not mechanically _____ the robot.

Safety reasoning

Why is stopped wheel motion not enough evidence that it is safe to approach?

Split arcade drive calculations

leftPower = forward + rotate;
rightPower = forward - rotate;

CASE	FORWARD	ROTATE	LEFT	RIGHT	PREDICTED MOTION
Stopped	0.00	0.00			
Forward	+0.60	0.00			
Reverse	-0.50	0.00			
Turn right	0.00	+0.40			
Forward right	+0.60	+0.20			
Full mix	+1.00	+1.00			

Normalization challenge

For forward = +1.00 and rotate = +1.00, the raw powers are left = _____ and right = _____.

Maximum magnitude = _____. Normalized left = _____. Normalized right = _____.

Why must both sides be divided by the same number?

Explain the turn

Why does forward = +0.60 and rotate = +0.20 produce a curve instead of a turn in place?

Intake trigger calculations

`intakePower = right_trigger - left_trigger;`

CASE	RIGHT TRIGGER	LEFT TRIGGER	INTAKE POWER	PREDICTED RESULT
Released	0.00	0.00		
Intake in	1.00	0.00		
Intake out	0.00	1.00		
Half speed in	0.50	0.00		
Equal triggers	0.75	0.75		

One value controls three devices

List the three devices that receive intakePower.

1. _____ 2. _____ 3. _____

Why is the right intake servo reversed?

A continuous-rotation servo uses power to control _____ and _____, not an exact position.

POWERING ON student check

Initial each step only when your team actually completes it.

STEP	SAFETY ACTION	INITIALS
1	Robot secured and moving parts clear of the surface	
2	Driver Station confirmed DISABLED	
3	Entire work area checked	
4	POWERING ON called loudly	
5	Team given time to step behind the boundary	
6	Hands, tools, wires, hair, clothing, and people visually clear	
7	Safety captain authorized INIT or START	

Wheels up observation record

TEST	PREDICTION	TELEMETRY	OBSERVED MOTION	PASS
Left drive				
Right drive				
Straight forward				
Rotate				
Intake inward				
Intake outward				
Equal triggers				
STOP and DISABLED				

Evidence based troubleshooting

Choose one unexpected result. Record evidence before proposing a change.

Unexpected result: _____

EVIDENCE	QUESTION	MY RECORD
Code	What calculation or command did the program contain?	
Configuration	Which configured name and device were used?	
Telemetry	What value did the Driver Station report?	
Observation	What did the physical mechanism actually do?	
Hypothesis	What single cause best fits all the evidence?	
Next safe test	What is the smallest test that could confirm it?	

Exit response

Explain how one gamepad controls the drivetrain, intake roller, and two feeder servos. Include one calculation and one safety or diagnostic check.

Student safety confirmation

I understand that I must press STOP, visually confirm DISABLED, and receive permission before approaching or adjusting the robot.

Student initials _____ Safety captain initials _____